Joseph Chombo Morocho

ML Engineer | MLOps | Python | SQL | FastAPI | MLflow | Databricks | Docker

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Professional Summary

ML Engineer focused on Python, SQL, and MLOps, with nearly 2 years in banking. Built ML validation pipelines on Databricks and PySpark that reduced manual execution by ∼60%, and a GenAI platform that cut validation time-to-market by ∼70%.

Technical Skills

- **Backend:** Python (advanced), FastAPI, REST APIs, input validation, logging, PostgreSQL, SQL Server, C#/.NET
- **MLOps:** MLflow (tracking, model registry, versioning), Databricks Feature Store, Databricks Workflows/Jobs, Unity Catalog, automated model validation, model monitoring (PSI, drift, calibration), batch inference
- **Data & SQL:** SQL (advanced), PySpark, Spark SQL, feature and data pipelines, data quality controls
- **Search & Retrieval:** embeddings, vector search (pgvector, Azure AI Search), semantic search, reranking, RAG, LLM evaluation and observability (Langfuse)
- **Infra & Cloud:** Docker, Kubernetes, GitHub Actions CI/CD, Azure (ADLS, OpenAI, AI Foundry), AWS, AI-assisted development (Claude Code)
- **Machine Learning:** scikit-learn, LightGBM, TensorFlow, classification, regression, feature engineering, model benchmarking
- **Languages:** Spanish (native), English (professional working proficiency, ICPNA certified)

Professional Experience

AI Engineer - Santander Consumer Peru (via Experis/ManpowerGroup) Sep 2026 – Present
Lima, Peru

- Build backend APIs that integrate internal databases, servers, and web applications for AI-powered services.
- Develop an LLM-based WhatsApp assistant that lets sales staff query business KPIs in natural language, with automatic chart generation.
- Design retrieval (RAG) flows and prompts for context-grounded, traceable answers; build internal agents with Microsoft Copilot Studio.

Machine Learning Engineer Intern - Banco de Crédito del Perú (BCP) Jan 2025 – Aug 2026
Lima, Peru

- Automated ML model validation workflows with Python, PySpark, Databricks, and Azure ADLS, reducing manual execution by ∼60%.
- Implemented experiment tracking, model versioning, and automated validations with MLflow, improving reproducibility and traceability across the model lifecycle.
- Built Feature Store components in Databricks and PySpark for feature generation, consolidation, and data quality controls.
- Designed model monitoring controls for discrimination, stability, and calibration (Gini, ROC-AUC, PSI).
- Built benchmarking components to compare Logistic Regression, tree ensembles, LightGBM, and neural networks for challenger model evaluation, working with data scientists and model owners to review results and agree on fixes.
- Built SoftIA, a GenAI platform (Python, Azure AI Foundry, OpenAI, Databricks, Streamlit) that automated model validation reports, cutting time-to-market by ∼70%; recognized two consecutive quarters for impact.
- Deployed an end-to-end RAG pipeline with Azure OpenAI, embeddings, semantic chunking, and reranking of retrieved results for relevance, with tracking of executions, tokens, and costs.

Full Stack Developer - Freelance Aug 2023 – Jan 2024
Lima, Peru

- Built an e-commerce platform with C#, .NET MVC, WebAssembly, Bootstrap, and SQL Server covering products, users, inventory, and purchase flows.

Key Projects

QnA End-to-End Service - github.com/JosephChomboM/qna-end-to-end-service
FastAPI, Docker, Kubernetes, MLflow, GitHub Actions

- Retrieval service covering corpus ingestion, embedding generation, vector search, and LLM answer generation behind a FastAPI backend.
- Evaluation pipeline tracked in MLflow, CI/CD with GitHub Actions, and deployment with Docker and Kubernetes.

MLOps Pipeline for Churn Prediction
Python, scikit-learn, MLflow, FastAPI, Docker

- End-to-end pipeline from data preparation and feature engineering to training and reproducible evaluation, with model comparison and versioning in MLflow.
- Batch inference and a containerized FastAPI serving API with input validation and logging.

AI Operational Workflow Assistant
Python, FastAPI, PostgreSQL, pgvector, Docker, Langfuse

- FastAPI backend for workflow state, auditing, and API integration that classifies and prioritizes operational requests with LLMs, business rules, and human-in-the-loop review.
- Vector search with pgvector and monitoring of latency, cost, and response quality with Langfuse.

Education

B.S. in Information Systems Engineering - Universidad San Ignacio de Loyola (USIL) Expected: Nov 2026

- Top 20% of class. Participant, USIL Quant Innovation Challenge 2023 and CITTIC 2023.

Certifications

- Microsoft Certified: SQL AI Developer Associate (DP-800) - Microsoft
- AWS Fundamentals Specialization - Amazon Web Services
- IBM Machine Learning Specialization - IBM
- Inferential Statistical Analysis with Python - University of Michigan
- SQL Specialization - Netzun